Finitely Additive Observable Laws and the Prokhorov Extension for Query Systems

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Abstract

The observational foundations program constructs a canonical probability space from a compatible family of observable laws, taking countable additivity of those laws as a hypothesis. This paper asks when that hypothesis can itself be derived. We show that for a countable Hausdorff query system with continuous refinement maps, any finitely compatible family of finitely additive normalized valuations satisfying a *surjective projective uniform tightness* condition admits a unique extension to a σ-additive probability measure on the canonical realization space. Countable additivity is not assumed locally — it is forced globally by the combination of projective compactness and sequential refinement structure.

The result is a Prokhorov-type extension theorem in the language of query systems. It complements the main extension theorem of *Observational Foundations of Probability* (which takes σ-additive marginals as input) by identifying conditions under which those marginals need not be assumed: they emerge from a tightness condition on the compatible family as a whole. The surjective projective uniform tightness condition is the query-system analogue of Prokhorov tightness, with the surjectivity of bonding maps on compact cores playing the role that single-space compactness plays in the classical theory.

1 Introduction

The foundational question. The observational program constructs probability from observation. Its central extension theorem (*Observational Foundations of Probability*, hereafter Paper 1) shows that a compatible family of observable laws $\{\nu_Q\}$ on a query system $(\mathcal{Q}, \preceq)$ determines a unique probability measure P on the canonical realization space Ω. The construction is a representation theorem: given that each query Q carries a countably additive law ν_Q, the global P is assembled from those laws.

The question this paper asks is foundationally prior:

> *Must one assume countable additivity at each query level, or can it be derived from the structure of the compatible family?*

This is Stopping Point 2 of the program: the assumption that each ν_Q is σ-additive is the fuel on which the extension theorem runs. The present paper identifies conditions under which that fuel can itself be derived — conditions that are properties of the *system* rather than of individual marginals.

The answer. We show that under a condition called *surjective projective uniform tightness* (SPUT), starting from only finitely additive normalized valuations $\{\ell_Q\}$ satisfying finite compatibility, one can derive a σ-additive global extension. The argument has two steps.

The first step is analytic: SPUT supplies compatible compact cores $\{K_Q\}$ whose inverse limit $K_\Omega \subseteq \Omega$ is compact and nonempty. Any decreasing sequence of cylinder sets with uniformly positive mass must meet K_Ω at each level; since K_Ω is compact, the intersection is nonempty, contradicting the assumption that the sequence decreases to $\emptyset$. This forces the cylinder premeasure μ_0 to be σ-subadditive, hence σ-additive by finite additivity and Carathéodory extension.

The second step is geometric: the extended measure lives on Ω rather than the ambient product space. A graph-support lemma shows that for each comparable pair $Q_1 \preceq Q_2$, the joint law of $(\mathrm{eval}_{Q_1}, \mathrm{eval}_{Q_2})$ under P is supported on the graph of $\pi_{Q_1}^{Q_2}$. For countable $\mathcal{Q}$, a union over countably many null defect sets gives $P(\Omega) = 1$.

What SPUT is. SPUT asks that for every $\varepsilon > 0$ one can find compact sets $K_Q \subset O_Q$ at each query level such that: (i) the bonding maps restrict to surjections between compact sets, $\pi_{Q_1}^{Q_2}(K_{Q_2}) = K_{Q_1}$; and (ii) the mass outside the compact core is uniformly small, $\ell_Q(O_Q \setminus K_Q) < \varepsilon$ for every Q. Condition (i) — surjectivity of the bonding maps on the compact cores — is the critical one. It ensures the compact inverse limit $K_\Omega = \varprojlim K_Q$ is nonempty with surjective projections, which is exactly what the compactness argument requires. Without surjectivity, K_Ω can be empty even when each K_Q is nonempty.

Comparison to classical Prokhorov. In classical Prokhorov tightness, one controls mass concentration in a single space: for every $\varepsilon > 0$, compact $K \subset X$ with $\mu_n(X \setminus K) < \varepsilon$. The query-system analogue must control concentration *projectively*, across all query levels simultaneously, and must ensure the compact sets assemble coherently in the inverse limit. SPUT is exactly this coherent projective tightness condition, with surjectivity of bonding maps playing the role that single-space compactness plays classically.

Relation to Paper 1. The two extension theorems are complementary:

	Paper 1 (realizability route)	This paper (tightness route)
Input measures	σ-additive ν_Q	finitely additive ℓ_Q
Key condition	realizability of Ω	surjective projective uniform tightness
Topology required	no	Hausdorff O_Q, continuous π
Conclusion	σ-additive P on Ω	σ-additive P on Ω

Paper 1 takes σ-additivity as given and derives a global measure from realizability. This paper takes only finite additivity and derives both σ-additivity and realizability from tightness. The price is topological structure on the query system.

Structure of the paper. Section 2 fixes notation and introduces finitely additive compatible families on query systems. Section 3 defines SPUT, motivates the surjectivity condition, and gives examples. Section 4 proves σ-subadditivity on closed cylinder sequences (the intermediate theorem). Section 5 extends to general Borel cylinders using Alexandrov regularity. Section 6 establishes concentration on Ω via the graph-support lemma. Section 7 assembles the full theorem and compares it to Paper 1. Section 8 discusses what the result achieves for the foundational program, what it does not yet achieve, and the open frontier.

2 Setup

Definition 1 (Query system)**.** A *query system* is a tuple $(\mathcal{Q}, \preceq, \{O_Q\}, \{\pi_{Q_1}^{Q_2}\})$ where:

- $(\mathcal{Q}, \preceq)$ is a countable directed preorder (the *query index set*);
- each $Q \in \mathcal{Q}$ is assigned a measurable outcome space $(O_Q, \mathcal{B}(O_Q))$;

- for each $Q_1 \preceq Q_2$, there is a measurable map $\pi_{Q_1}^{Q_2} : O_{Q_2} \to O_{Q_1}$ (the *refinement map*) satisfying $\pi_{Q_1}^{Q_1} = \mathrm{id}$ and $\pi_{Q_1}^{Q_3} = \pi_{Q_1}^{Q_2} \circ \pi_{Q_2}^{Q_3}$ whenever $Q_1 \preceq Q_2 \preceq Q_3$.

The system is *sequentially upper-directed* if for every countable family $\{Q_n\}_{n\geq 1} \subset \mathcal{Q}$ there exists $Q_* \in \mathcal{Q}$ with $Q_* \succeq Q_n$ for all n.

Remark 1 (Topological enrichment)**.** Throughout this paper the outcome spaces O_Q are assumed to be Hausdorff topological spaces with $\mathcal{B}(O_Q)$ the Borel σ-algebra, and the refinement maps $\pi_{Q_1}^{Q_2}$ are assumed to be continuous. These topological conditions are not required by Paper 1 and are the price paid for starting from only finitely additive marginals.

Definition 2 (Canonical realization space)**.** The *canonical realization space* is the projective limit

$$\Omega := \varprojlim_{Q\in\mathcal{Q}} O_Q = \left\{ \omega \in \prod_{Q\in\mathcal{Q}} O_Q \;\middle|\; \pi_{Q_1}^{Q_2}(\omega_{Q_2}) = \omega_{Q_1} \;\forall\, Q_1 \preceq Q_2 \right\},$$

equipped with evaluation maps $\mathrm{eval}_Q : \Omega \to O_Q$, $\omega \mapsto \omega_Q$.

Since each $\pi_{Q_1}^{Q_2}$ is continuous and each O_Q is Hausdorff, each coherence constraint set $\{\omega : \pi_{Q_1}^{Q_2}(\omega_{Q_2}) = \omega_{Q_1}\}$ is closed in the product. Since $\mathcal{Q}$ is countable, Ω is a countable intersection of closed sets, hence closed in $\prod_Q O_Q$.

The *cylinder algebra* $\mathcal{C}$ is generated by sets $\mathrm{Cyl}(Q, A) := \mathrm{eval}_Q^{-1}(A)$ for $Q \in \mathcal{Q}$, $A \in \mathcal{B}(O_Q)$, and the *observable σ-field* is $\sigma(\mathcal{Q}) := \sigma(\mathcal{C})$.

Definition 3 (Finitely compatible family)**.** A family $\{\ell_Q\}_{Q\in\mathcal{Q}}$ of normalized valuations $\ell_Q : \mathcal{B}(O_Q) \to [0, 1]$ (with $\ell_Q(O_Q) = 1$) is *finitely compatible* if for every $Q_1 \preceq Q_2$,

$$(\pi_{Q_1}^{Q_2})_{\#}\ell_{Q_2} = \ell_{Q_1},$$

i.e. $\ell_{Q_1}(A) = \ell_{Q_2}((\pi_{Q_1}^{Q_2})^{-1}(A))$ for all $A \in \mathcal{B}(O_{Q_1})$. Each ℓ_Q is assumed to be finitely additive (but not necessarily σ-additive).

Remark 2 (The cylinder premeasure)**.** Given a finitely compatible family $\{\ell_Q\}$, define the *cylinder premeasure* $\mu_0 : \mathcal{C} \to [0, 1]$ by

$$\mu_0(\mathrm{Cyl}(Q, A)) := \ell_Q(A).$$

Finite compatibility ensures μ_0 is well-defined (independent of which query level is used to represent a given cylinder) and finitely additive on $\mathcal{C}$. The central question is when μ_0 is σ-additive, so that Carathéodory's theorem extends it to a probability measure on $\sigma(\mathcal{Q})$.

3 Surjective projective uniform tightness

Definition 4 (Surjective projective uniform tightness)**.** A finitely compatible family $\{\ell_Q\}$ is *surjectively projectively uniformly tight* (SPUT) if for every $\varepsilon > 0$ there exists a system $\{K_Q\}_{Q\in\mathcal{Q}}$ of compact sets $K_Q \subset O_Q$ such that:

(i) *Surjective bonding:* $\pi_{Q_1}^{Q_2}(K_{Q_2}) = K_{Q_1}$ for all $Q_1 \preceq Q_2$;

(ii) *Uniform mass control:* $\ell_Q(O_Q \setminus K_Q) < \varepsilon$ for every $Q \in \mathcal{Q}$.

The two conditions play distinct roles. Condition (ii) ensures that the compact cores capture most of the mass at every query level. Condition (i) ensures the compact cores assemble coherently across levels, so their inverse limit is nonempty.

Remark 3 (Why surjectivity is necessary)**.** If condition (i) required only $\pi_{Q_1}^{Q_2}(K_{Q_2}) \subseteq K_{Q_1}$ (containment rather than equality), the inverse limit $K_\Omega := \Omega \cap \prod_Q K_Q$ could be empty even when each K_Q is nonempty. The classical example: let $O_Q = [0,1]$, $K_Q = [0, 1 - 1/Q]$, with refinement maps the identity. Then each K_Q is nonempty and the bonding maps satisfy containment, but $\bigcap_Q K_Q = \emptyset$. Surjectivity — $\pi(K_{Q_2}) = K_{Q_1}$ exactly — rules out such "shrinking core" phenomena and guarantees $K_\Omega \neq \emptyset$ with surjective projections (Proposition 1).

Remark 4 (Comparison to classical Prokhorov tightness)**.** Classical tightness for a family of measures $\{\mu_n\}$ on a single metric space X asks: for every $\varepsilon > 0$, a compact $K \subset X$ with $\mu_n(K) \geq 1 - \varepsilon$ for all n. SPUT is the projective analogue: compact sets must be chosen at every query level simultaneously, and they must assemble into a compact subset of Ω via the surjectivity condition. The surjectivity of bonding maps is the multi-space analogue of the fact that K is a single space (automatically surjective over itself) in the classical case.

Remark 5 (When SPUT is achievable)**.** SPUT holds automatically when the refinement maps $\pi_{Q_1}^{Q_2}$ are themselves surjective. In that case, given any compact K_{Q_*} at a fine level Q_*, one constructs compatible compact cores by setting $K_Q := \pi_Q^{Q_*}(K_{Q_*})$ for each $Q \preceq Q_*$; surjectivity of the bonding maps ensures $\pi_{Q_1}^{Q_2}(K_{Q_2}) = K_{Q_1}$. Thus the delay query systems of *Observational Probability from Delay Queries* (where the refinement maps are coordinate projections, which are surjective) satisfy SPUT whenever the individual marginals are tight in the classical sense.

Definition 5 (Compact core)**.** Given a SPUT system $\{K_Q\}$, the *compact core* of Ω is

$$K_\Omega := \Omega \cap \prod_{Q \in \mathcal{Q}} K_Q = \{\omega \in \Omega \mid \omega_Q \in K_Q \ \forall Q \in \mathcal{Q}\}.$$

Proposition 1 (Properties of the compact core)**.** *Under SPUT:*

(i) K_Ω is compact.

(ii) K_Ω is nonempty, and $\mathrm{eval}_Q(K_\Omega) = K_Q$ *for every $Q \in \mathcal{Q}$.*

Proof. (i) The product $\prod_Q K_Q$ is compact by Tychonoff's theorem. Since Ω is closed in $\prod_Q O_Q$ (as each coherence constraint is a closed condition), $K_\Omega = \Omega \cap \prod_Q K_Q$ is a closed subset of a compact space, hence compact.

(ii) The system $(K_Q, \pi_{Q_1}^{Q_2}|_{K_{Q_2}})$ is an inverse system of nonempty compact Hausdorff spaces. By condition (i) of SPUT, the bonding maps $\pi_{Q_1}^{Q_2}|_{K_{Q_2}} : K_{Q_2} \to K_{Q_1}$ are surjective. By the *inverse limit theorem* for compact Hausdorff spaces with surjective bonding maps over a directed index set, $K_\Omega = \varprojlim K_Q$ is nonempty and each projection $\mathrm{eval}_Q : K_\Omega \to K_Q$ is surjective. □

Remark 6 (The inverse limit theorem)**.** Proposition 1(ii) uses the classical result that an inverse system of nonempty compact Hausdorff spaces with surjective bonding maps has a nonempty inverse limit with surjective projections. This follows by expressing K_Ω as an intersection in the compact product $\prod_Q K_Q$, where each finite sub-intersection is nonempty by surjectivity, and applying the finite intersection property. The directed-set version of this argument is standard but is not yet available in Mathlib 4, and constitutes the main remaining gap in the Lean formalization.

4 The intermediate theorem: closed cylinders

The key analytic step is to show that the cylinder premeasure μ_0 is σ-subadditive on sequences of closed cylinders.

Theorem 1 (Closed cylinder σ-subadditivity)**.** *Let $(\mathcal{Q}, \preceq)$ be a countable sequentially upper-directed query system with Hausdorff outcome spaces and continuous refinement maps. Let $\{\ell_Q\}$ be a finitely compatible SPUT family. If $C_N = \mathrm{Cyl}(Q_N, F_N) \downarrow \emptyset$ is a decreasing sequence of cylinder sets with each $F_N \subseteq O_{Q_N}$ closed, then*

$$\mu_0(C_N) \to 0.$$

Proof. Suppose for contradiction that $\mu_0(C_N) \geq \varepsilon > 0$ for all N.

Step 1 (Compact core). Apply SPUT with parameter $\varepsilon/2$: obtain compact sets $\{K_Q\}$ with surjective bonding maps and $\ell_Q(O_Q \setminus K_Q) < \varepsilon/2$ for all Q.

Step 2 (K_Ω is compact and nonempty with surjective projections). By Proposition 1, K_Ω is compact and $\mathrm{eval}_{Q_N}(K_\Omega) = K_{Q_N}$ for every N.

Step 3 (Each $C_N \cap K_\Omega$ is nonempty). Since $\ell_{Q_N}(F_N) = \mu_0(C_N) \geq \varepsilon$ and $\ell_{Q_N}(K_{Q_N}) \geq 1 - \varepsilon/2$, we have

$$\ell_{Q_N}(F_N \cap K_{Q_N}) \geq \varepsilon - \varepsilon/2 = \varepsilon/2 > 0.$$

In particular $F_N \cap K_{Q_N} \neq \emptyset$. Since $\mathrm{eval}_{Q_N} : K_\Omega \to K_{Q_N}$ is surjective, there exists $\omega \in K_\Omega$ with $\mathrm{eval}_{Q_N}(\omega) \in F_N \cap K_{Q_N}$, so $\omega \in C_N \cap K_\Omega$.

Step 4 (Each $C_N \cap K_\Omega$ is closed in K_Ω). $C_N = \mathrm{eval}_{Q_N}^{-1}(F_N)$ where F_N is closed and eval_{Q_N} is continuous, so C_N is closed in Ω and $C_N \cap K_\Omega$ is closed in K_Ω.

Step 5 (Contradiction by compactness). The sets $\{C_N \cap K_\Omega\}$ form a decreasing sequence of nonempty closed subsets of the compact space K_Ω. By the finite intersection property,

$$\bigcap_N (C_N \cap K_\Omega) \neq \emptyset.$$

But $\bigcap_N C_N = \emptyset$ (since $C_N \downarrow \emptyset$), so $\bigcap_N (C_N \cap K_\Omega) \subseteq \bigcap_N C_N = \emptyset$. Contradiction. $\square$

5 Extension to Borel cylinders

The intermediate theorem covers closed cylinders. We now extend to all Borel cylinders. The key observation is that no new hypotheses are needed: inner regularity on each compact core is automatic.

Lemma 1 (Alexandrov regularity is free)**.** *Let K be a compact Hausdorff space and ν a finite Borel measure on K. Then ν is inner regular: for every Borel $A \subseteq K$ and $\eta > 0$, there exists closed $F \subseteq A$ with $\nu(A \setminus F) < \eta$.*

This is Alexandrov's theorem; it requires only that K is compact Hausdorff, which follows from SPUT.

Theorem 2 (Borel cylinder σ-subadditivity)**.** *Under the hypotheses of Theorem 1, if $C_N = \mathrm{Cyl}(Q_N, A_N) \downarrow \emptyset$ is a decreasing sequence of Borel cylinder sets, then $\mu_0(C_N) \to 0$.*

Proof. Suppose $\mu_0(C_N) \geq \varepsilon > 0$ for all N. Set $\varepsilon' := \varepsilon/4$.

Apply SPUT with parameter ε', obtaining compact cores $\{K_Q\}$ with $\ell_Q(O_Q \setminus K_Q) < \varepsilon'$ for all Q.

For each N, apply Alexandrov regularity (Lemma 1) to $\ell_{Q_N}|_{K_{Q_N}}$ with tolerance $\varepsilon'/2^N$: find closed $F_N \subseteq A_N \cap K_{Q_N}$ (closed in O_{Q_N}, since K_{Q_N} is compact in the Hausdorff space O_{Q_N}) with

$$\ell_{Q_N}(A_N \cap K_{Q_N}) - \ell_{Q_N}(F_N) < \varepsilon'/2^N.$$

Define closed cylinders $D_N := \mathrm{Cyl}(Q_N, F_N)$. Then

$$\mu_0(D_N) = \ell_{Q_N}(F_N) \geq \ell_{Q_N}(A_N) - \varepsilon' - \varepsilon'/2^N \geq \varepsilon - 2\varepsilon' = \varepsilon/2 > 0.$$

Let $E_N := \bigcap_{k=1}^N D_k$. Since the system is sequentially upper-directed, finite intersections of closed cylinders are again closed cylinders (each pair (D_i, D_j) has a common refinement $Q_{ij} \succeq Q_i, Q_j$, giving $D_i \cap D_j = \mathrm{Cyl}(Q_{ij}, (\pi^{Q_{ij}}_{Q_i})^{-1}(F_i) \cap (\pi^{Q_{ij}}_{Q_j})^{-1}(F_j))$, which is a closed cylinder). So each E_N is a closed cylinder.

Moreover, since C_N is decreasing and $D_k \subseteq C_k$:

$$\mu_0(E_N) \geq \mu_0(C_N) - \sum_{k=1}^{N} \mu_0(C_k \setminus D_k) \geq \varepsilon - \sum_{k=1}^{\infty} 2\varepsilon'/2^k = \varepsilon - 2\varepsilon'(1+1) = \varepsilon/2 > 0.$$

Since $E_N \subseteq C_N$ and $C_N \downarrow \emptyset$, we have $E_N \downarrow \emptyset$ with each $\mu_0(E_N) \geq \varepsilon/2 > 0$, contradicting Theorem 1. □

Corollary 1 (σ-additivity of μ_0)**.** *Under the above hypotheses, the cylinder premeasure μ_0 is σ-additive on $\mathcal{C}$. By Carathéodory's extension theorem, μ_0 extends uniquely to a σ-additive probability measure on $\sigma(\mathcal{Q})$ over the ambient product $\prod_Q O_Q$.*

6 Concentration on Ω

The Carathéodory extension lives on the ambient product $\prod_Q O_Q$. We must show it concentrates on the projective limit Ω.

Lemma 2 (Graph support)**.** *Let P be the σ-additive extension of μ_0 to $\prod_Q O_Q$. For each $Q_1 \preceq Q_2$, the joint law of* $(\mathrm{eval}_{Q_1}, \mathrm{eval}_{Q_2})$ *under P is supported on the graph*

$$G_{Q_1,Q_2} := \big\{(a,b) \in O_{Q_1} \times O_{Q_2} \;\big|\; a = \pi^{Q_2}_{Q_1}(b)\big\}.$$

Proof. It suffices to verify on measurable rectangles, which generate $\mathcal{B}(O_{Q_1}) \otimes \mathcal{B}(O_{Q_2})$. For $A \in \mathcal{B}(O_{Q_1})$ and $B \in \mathcal{B}(O_{Q_2})$:

$$\begin{aligned} P\big(\mathrm{eval}^{-1}_{Q_1}(A) \cap \mathrm{eval}^{-1}_{Q_2}(B)\big) &= \mu_0\big(\mathrm{Cyl}(Q_2, (\pi^{Q_2}_{Q_1})^{-1}(A) \cap B)\big) \\ &= \ell_{Q_2}\big((\pi^{Q_2}_{Q_1})^{-1}(A) \cap B\big), \end{aligned}$$

which equals $(\mathrm{graph}_\pi)_\# \ell_{Q_2}(A \times B)$ where $\mathrm{graph}_\pi : b \mapsto (\pi^{Q_2}_{Q_1}(b), b)$. The measure $(\mathrm{graph}_\pi)_\# \ell_{Q_2}$ is entirely supported on G_{Q_1,Q_2}.

The graph G_{Q_1,Q_2} is Borel: it is the preimage of the diagonal $\Delta_{O_{Q_1}}$ (closed in Hausdorff $O_{Q_1} \times O_{Q_1}$) under the continuous map $(a,b) \mapsto (a, \pi^{Q_2}_{Q_1}(b))$. Hence G_{Q_1,Q_2} is closed, so the statement is well-posed. □

Corollary 2. *For each $Q_1 \preceq Q_2$, defining the defect set $D_{Q_1,Q_2} := \{\omega : \omega_{Q_1} \neq \pi^{Q_2}_{Q_1}(\omega_{Q_2})\}$, we have $P(D_{Q_1,Q_2}) = 0$.*

Theorem 3 (Concentration on Ω)**.** *$P(\Omega) = 1$.*

Proof. The complement decomposes as $\prod_Q O_Q \setminus \Omega = \bigcup_{Q_1 \preceq Q_2} D_{Q_1,Q_2}$. Since $\mathcal{Q}$ is countable, the preorder has countably many comparable pairs. By Corollary 2, each D_{Q_1,Q_2} is a P-null set. A countable union of null sets is null, so $P(\prod_Q O_Q \setminus \Omega) = 0$. □

7 The main theorem

Theorem 4 (Prokhorov extension for query systems)**.** *Let $(\mathcal{Q}, \preceq)$ be a countable sequentially upper-directed query system in which each outcome space O_Q is a Hausdorff topological* *space, $\mathcal{B}(O_Q)$ is its Borel σ-algebra, and each refinement map $\pi^{Q_2}_{Q_1} : O_{Q_2} \to O_{Q_1}$ is continuous. Let $\{\ell_Q\}_{Q\in\mathcal{Q}}$ be a finitely compatible family of finitely additive normalized valuations on $\{(O_Q, \mathcal{B}(O_Q))\}$.*

If $\{\ell_Q\}$ is surjectively projectively uniformly tight (Definition 4), then there exists a unique σ-additive probability measure P on $(\Omega, \sigma(\mathcal{Q}))$ satisfying

$$(\mathrm{eval}_Q)_\# P = \ell_Q$$

for every $Q \in \mathcal{Q}$. In particular, each ℓ_Q is σ-additive.

Proof. σ-additivity of μ_0. By Theorem 2, μ_0 satisfies continuity from above at $\emptyset$ on all Borel cylinders. Since μ_0 is already finitely additive, it is σ-additive on $\mathcal{C}$.

Carathéodory extension. By Carathéodory's theorem, μ_0 extends uniquely to a σ-additive probability measure, still called P, on $\sigma(\mathcal{C})$ over $\prod_Q O_Q$.

Concentration. By Theorem 3, $P(\Omega) = 1$. The restriction $P|_\Omega$ is the required measure on $(\Omega, \sigma(\mathcal{Q}))$.

Marginal recovery. For any $Q \in \mathcal{Q}$ and $A \in \mathcal{B}(O_Q)$: $P(\mathrm{eval}_Q^{-1}(A)) = \mu_0(\mathrm{Cyl}(Q, A)) = \ell_Q(A)$.

Uniqueness. The observable σ-field $\sigma(\mathcal{Q})$ is generated by the cylinder algebra $\mathcal{C}$. Any two σ-additive probability measures on $(\Omega, \sigma(\mathcal{Q}))$ with the same ℓ_Q-marginals agree on $\mathcal{C}$ and hence on $\sigma(\mathcal{Q})$. □

Remark 7 (Induced σ-additivity)**.** The last sentence of Theorem 4 is the key point for the foundational program: the ℓ_Q were assumed only finitely additive, but the theorem produces a σ-additive global measure whose marginals are exactly the ℓ_Q. In particular, each $\ell_Q = (\mathrm{eval}_Q)_\# P$ is σ-additive (as a pushforward of a σ-additive measure). Countable additivity is not an input — it is an output.

Remark 8 (Hypothesis audit)**.** Every hypothesis in Theorem 4 is used, and each at exactly one point:

Hypothesis	Where used	Can it be weakened?
$\mathcal{Q}$ countable	Concentration (§6)	Countable determining subsystem suffices
Sequentially upper-directed	Borel extension (§5)	Needed for closed-cylinder intersections
Hausdorff O_Q	Ω closed; graphs Borel	Essential
Continuous π	Ω closed; cylinders closed	Essential
Surjective bonding on K_Q	$K_\Omega \neq \emptyset$	Cannot weaken to containment
Compact K_Q	Tychonoff; FIP; Alexandrov	Essential
$\ell_Q(O_Q \setminus K_Q) < \varepsilon$	Mass estimates	Essential
Finite compatibility	μ_0 well-defined	Essential

8 Discussion

What the theorem achieves. Theorem 4 addresses Stopping Point 2 of the observational program: it shows that σ-additivity need not be assumed at each query level. Instead, it is a global consequence of two structural conditions: finite compatibility (the marginals cohere under refinement) and SPUT (the mass concentrates projectively on compatible compact cores). These conditions are properties of the *system* of observable laws, not of individual marginals.

The result is a strict weakening of the input to Paper 1: Paper 1 assumes σ-additive marginals and realizability, while this paper needs only finitely additive marginals and SPUT. The conclusion is the same: a σ-additive global measure on $(\Omega, \sigma(\mathcal{Q}))$.

What SPUT is buying. SPUT is a condition about how observable mass concentrates as one moves between query levels. It says, in operational terms: the things the observer considers *likely* at fine resolutions assemble coherently into things that are likely at coarser resolutions, and this assembly is non-degenerate (surjectivity). This is a natural condition on a query system arising from a physical process: if the system concentrates mass on a compact region at every observable resolution, that region is observable at all resolutions jointly.

What it costs. SPUT requires topological structure: Hausdorff spaces and continuous refinement maps. This is the price for starting from finitely additive marginals. Paper 1 operates without this topology, at the cost of requiring σ-additivity as an input. The two theorems occupy adjacent positions in the space of hypotheses, with this paper trading topology for weaker input measures.

The deeper question. Stopping Point 2 in its strongest form asks whether σ-additivity can be forced from *purely combinatorial* conditions on the query system, without topological assumptions. Theorem 4 does not achieve this: it replaces the assumption of σ-additivity with the assumption of Hausdorff topology plus SPUT. The question is whether SPUT itself can be derived from something more primitive.

One route: if the query system arises from a physical measurement process with compact outcome spaces (which is natural for bounded sensors), then the O_Q are automatically compact Hausdorff and SPUT reduces to the surjectivity of bonding maps on the full spaces — which holds whenever the refinement maps are surjective. Under these conditions, Theorem 4 applies with no additional hypotheses beyond finite compatibility and surjective refinement maps. This is the fully observational version: the query system architecture itself forces σ-additivity.

Open problems.

1. *Surjectivity of refinement maps as the primitive.* Can the SPUT condition be replaced, in the compact outcome space setting, by surjectivity of the refinement maps alone? This would remove the "for every $\varepsilon > 0$" quantifier from the hypothesis, leaving a purely structural condition.

2. *The uncountable case.* Concentration (Theorem 3) uses countability of $\mathcal{Q}$ to take a countable union of null defect sets. For uncountable $\mathcal{Q}$, this argument fails. The correct generalization appears to be a condition on determining subsystems: if $\mathcal{Q}$ admits a countable determining subsystem $\mathcal{Q}_0$ (coherence on $\mathcal{Q}_0$ forces coherence on all of $\mathcal{Q}$), then the countable argument applies. Characterizing when such subsystems exist is the open problem.

3. *Topology-free route.* Theorem 4 requires Hausdorff topology and continuity of refinement maps. A topology-free version would need to replace Alexandrov's theorem (automatic inner regularity on compact Hausdorff spaces) with a purely measure-theoretic substitute. Whether this is possible — and what structural condition on the query system it would require — remains open.

4. *Lean formalization.* The main outstanding gap in the Lean 4/Mathlib formalization is the directed-set form of the inverse limit theorem (Proposition 1(ii)). Mathlib 4 does not yet contain this result in the form needed. All other steps of the proof have been formalized or have clear paths to formalization once this gap is closed.

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